ASWIN BABU K M

DevOps Engineer

Madurai, India | *+91 9342489851* | *ashwinbabu769@gmail.com* | linkedin.com/in/aswin-babu-km | https://github.com/Aswinbabu5 | https://portfolio-nlb1.vercel.app

PROFESSIONAL SUMMARY

DevOps Engineer with hands-on training experience in Linux production support, infrastructure monitoring, and troubleshooting distributed systems at Zoho. Investigated production incidents through log analysis, and worked with Linux internals, eBPF, networking, and automation. Strong backend development knowledge in Java, Python, Spring Boot, and REST API development.

TECHNICAL SKILLS

DevOps & Systems: Linux Administration (RHEL/Ubuntu), Shell Scripting, Docker, Kubernetes, Ansible, jenksin, AWS.

Monitoring: Grafana, Prometheus.

Networking: TCP/IP, Packet Processing, eBPF.

Programming language: C/C++, Java, Bash, Python.

Backend & APIs: Spring Boot, REST APIs, SQL.

Tools & Platforms: Git, Postman, VSCode.

Concepts: Linux, Data Structures, OOP.

PROFESSIONAL EXPERIENCE

ZOHO CORPORATION PVT LTD — *Chengalpattu, India*

Technical Trainee

September 2025 – March 2026

- Monitored production infrastructure within local DC, assisting in the investigation of production incidents through log analysis under mentor guidance, and gained hands-on experience collaborating with engineering teams on infrastructure troubleshooting.
- Diagnosed and resolved a production incident where a disconnection between an internal binary service and a Linux network service caused a complete halt in processing flow; identified the root cause through script-level investigation and restored functionality by updating service connection configuration.
- Built a proof-of-concept feature flag management tool (Struts, Servlets, MySQL, XML) enabling teams to create, delete, update, view and toggle feature flag status, improving visibility and scalability into flag states and simplifying manual enable/disable operations.

Internship

July 2025 – August 2025

- Completed the assigned tasks comparing and benchmarking select, poll, epoll, and io_uring system calls, writing hands-on programs to evaluate asynchronous I/O approaches and their performance trade-offs.
- Implemented an eBPF program using an XDP pass hook to inspect packet flow through the machine, gaining hands-on experience with low-level packet processing and network interface operations.
- Conducted User-level experiments involving context switching, system call execution, and strengthening system-level debugging capabilities.

EDUCATION

B.Tech in Information Technology
K.L.N College of Engineering, *Sivagangai* *November 2022 – April 2026*

Higher Secondary Certificate (HSC)
Sourashtra Boys Higher Secondary School, *Madurai* *September 2021 – May 2022*

Secondary School Leaving Certificate (SSLC)
Seventh-Day Adventist Matriculation Hr Sec School, *Madurai* *June 2019 – April 2020*

PROJECTS

The Judicial Assistant — Jubilee
July 2025 – December 2025

- Developed an ML-powered automation pipeline in Python to assist users in retrieving and analyzing legal case records, demonstrating data processing and backend scripting capabilities.
- Applied NLP techniques for intelligent search and filtering of structured legal datasets, improving information retrieval efficiency.
- Integrated voice input, text-to-speech (Read Aloud), and translation features to make case information accessible through audio output.
- ***Technologies*:** Python, NLP, Scikit-learn, Pandas, Machine Learning
- https://jublie.streamlit.app

Smart Air Quality Monitoring for Asthma Patients
January 2026 – April 2026

- Built an IoT-based air quality monitoring system using Raspberry Pi and ESP32.
- Integrated four environmental sensors.
- Implemented Wi-Fi communication for real-time data transfer.
- Developed Python scripts for data collection and monitoring.
- Achieved real-time transmission with latency under 500 ms.
- ***Technologies*:** Python, Flutter, Raspberry pi 4, ESP 32, C++

Developer Dashboard
July 2026 – August 2026

- Developed a responsive developer dashboard with project tracking, Kanban board, analytics, and team management.
- Built reusable React components and responsive user interfaces.
- Applied routing, component-based architecture, and state management principles.
- ***Technologies*:** React.js, LocalStorage, BarCharts, Lucide-react
- https://developer-dashboard-tan.vercel.app/

CERTIFICATIONS

- Backend Developer — Meta (Coursera)
- Python Programming — SSI Institution
- C Programming Training — IIT Spoken Tutorial
- Smart Traffic Management System — Smart India Hackathon (PRE-SIH 2024)
- AI Tutor Chatbot — 24-Hour Hackathon, Ramakrishna College of Engineering, Trichy